

Writing Macros II

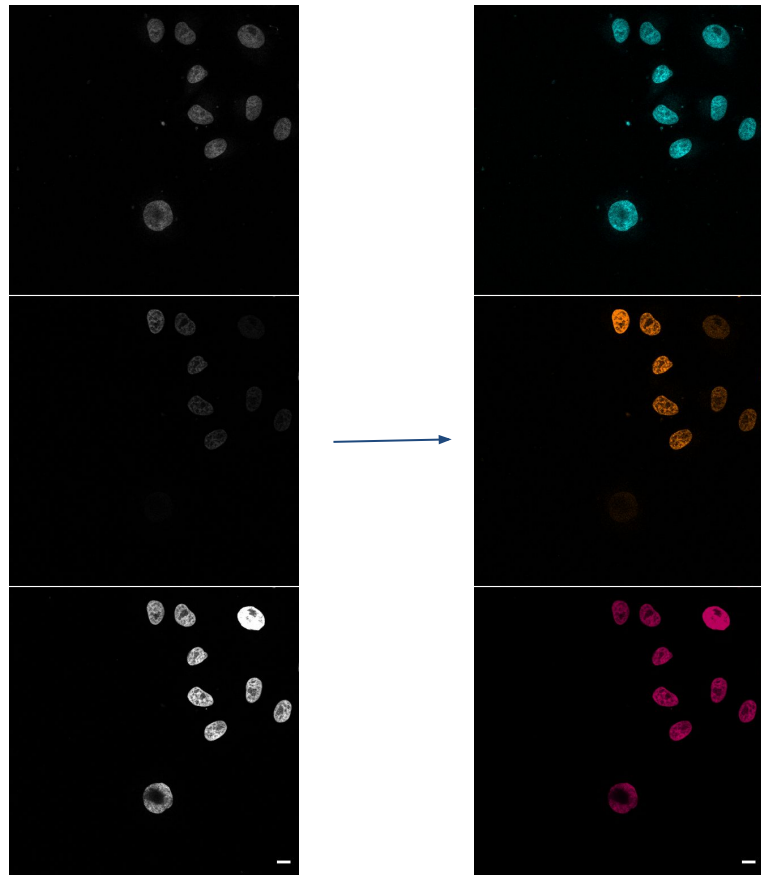
Image Processing & Analysis for Life Scientists

Olivier Burri, Romain Guiet & Arne Seitz

Writing your own Macro

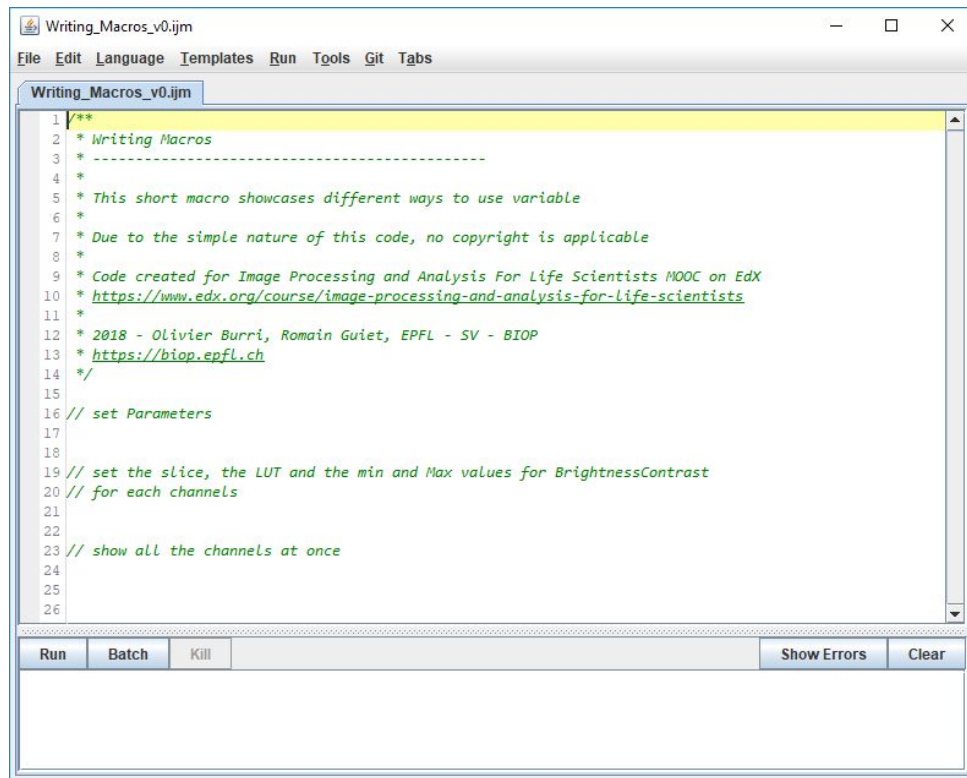
Suppose we have a series of images on which we want to:

1. Apply specific lookup tables
2. Adjust the brightness & contrast for each channel
3. Show all channels at once



Writing Macros from Scratch

Begin by laying out your workflow
using **comments**



```
1 /**
2  * Writing Macros
3  * -----
4  *
5  * This short macro showcases different ways to use variable
6  *
7  * Due to the simple nature of this code, no copyright is applicable
8  *
9  * Code created for Image Processing and Analysis For Life Scientists MOOC on EdX
10 * https://www.edx.org/course/image-processing-and-analysis-for-life-scientists
11 *
12 * 2018 - Olivier Burri, Romain Guet, EPFL - SV - BIOP
13 * https://biop.epfl.ch
14 */
15
16 // set Parameters
17
18
19 // set the slice, the LUT and the min and Max values for BrightnessContrast
20 // for each channels
21
22
23 // show all the channels at once
24
25
26
```

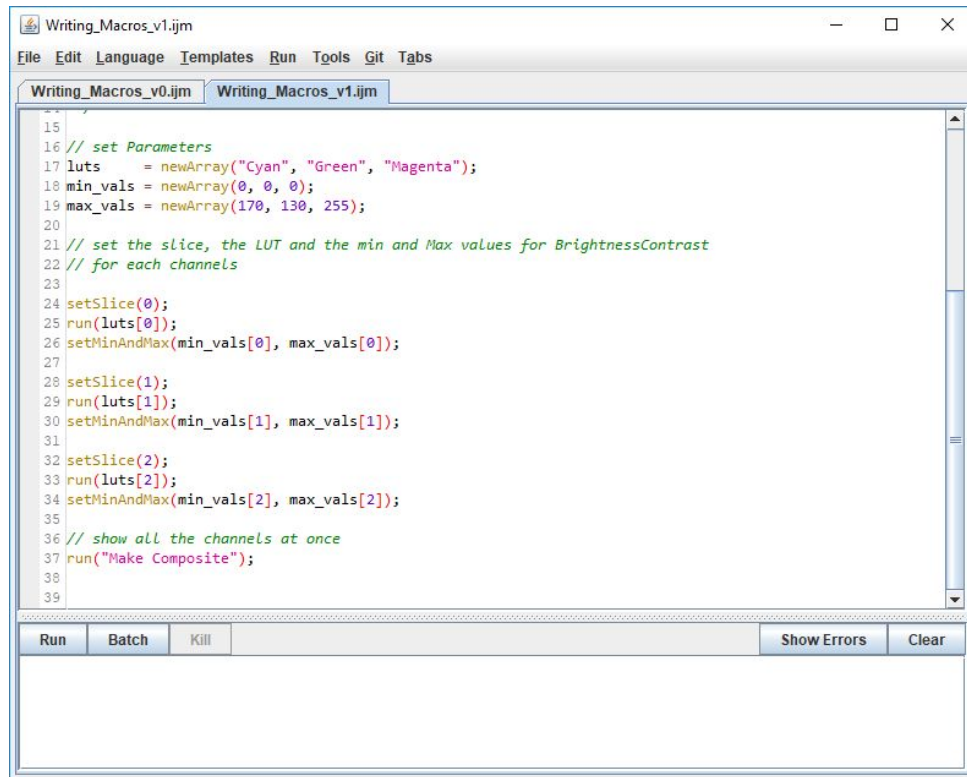
Run Batch Kill Show Errors Clear

Writing Macros from Scratch

Write your code between the comments. This code is very similar the the first one you made in the previous slides.

This code already works.

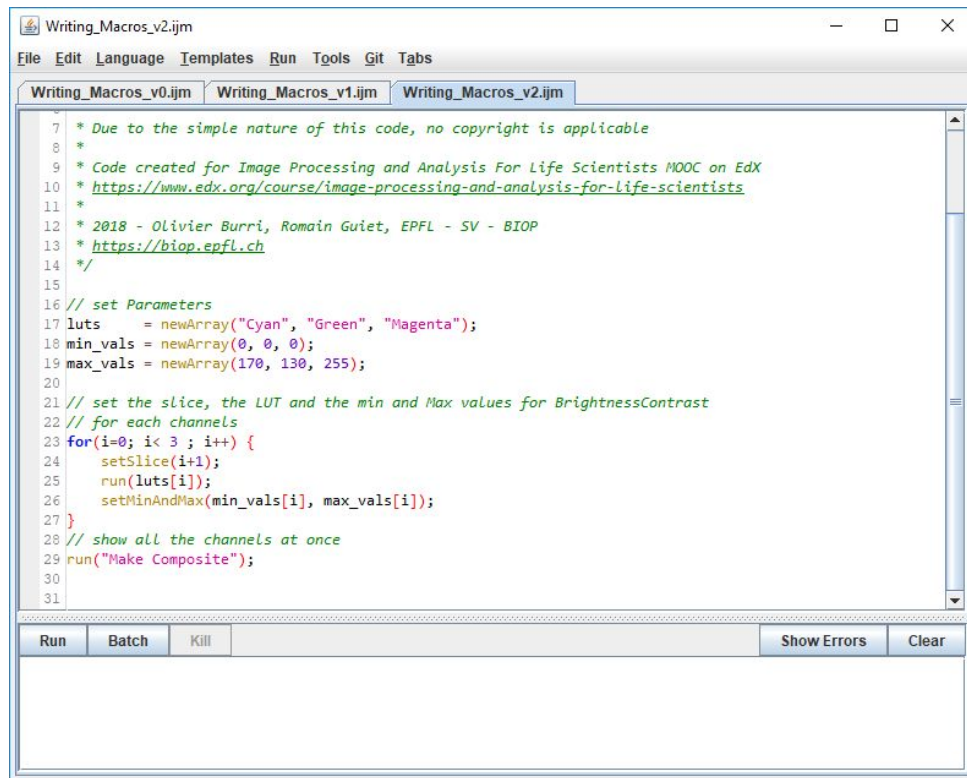
Now you can make it more generic



```
Writing_Macros_v1.ijm
File Edit Language Templates Run Tools Git Tabs
Writing_Macros_v0.ijm Writing_Macros_v1.ijm
15
16 // set Parameters
17 luts = newArray("Cyan", "Green", "Magenta");
18 min_vals = newArray(0, 0, 0);
19 max_vals = newArray(170, 130, 255);
20
21 // set the slice, the LUT and the min and Max values for BrightnessContrast
22 // for each channels
23
24 setSlice(0);
25 run(luts[0]);
26 setMinAndMax(min_vals[0], max_vals[0]);
27
28 setSlice(1);
29 run(luts[1]);
30 setMinAndMax(min_vals[1], max_vals[1]);
31
32 setSlice(2);
33 run(luts[2]);
34 setMinAndMax(min_vals[2], max_vals[2]);
35
36 // show all the channels at once
37 run("Make Composite");
38
39
Run Batch Kill Show Errors Clear
```

Writing Macros from Scratch

The repeating parts of the code have been placed in a **for** Loop

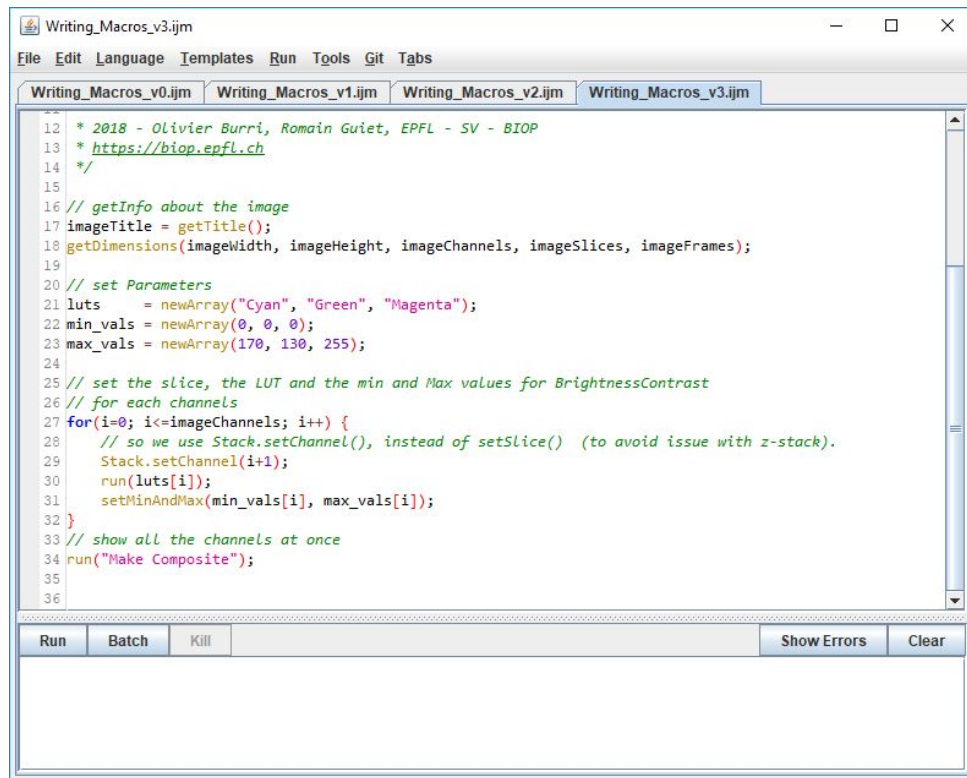


```
Writing_Macros_v2.ijm
File Edit Language Templates Run Tools Git Tabs
Writing_Macros_v0.ijm Writing_Macros_v1.ijm Writing_Macros_v2.ijm
7  * Due to the simple nature of this code, no copyright is applicable
8  *
9  * Code created for Image Processing and Analysis For Life Scientists MOOC on EdX
10 * https://www.edx.org/course/image-processing-and-analysis-for-life-scientists
11 *
12 * 2018 - Olivier Burri, Romain Guet, EPFL - SV - BIOP
13 * https://biop.epfl.ch
14 */
15
16 // set Parameters
17 luts    = newArray("Cyan", "Green", "Magenta");
18 min_vals = newArray(0, 0, 0);
19 max_vals = newArray(170, 130, 255);
20
21 // set the slice, the LUT and the min and Max values for BrightnessContrast
22 // for each channels
23 for(i=0; i< 3 ; i++) {
24     setSlice(i+1);
25     run(luts[i]);
26     setMinAndMax(min_vals[i], max_vals[i]);
27 }
28 // show all the channels at once
29 run("Make Composite");
30
31
Run Batch KIII Show Errors Clear
```

Writing Macros from Scratch

To account for the possibility of multiple channels:

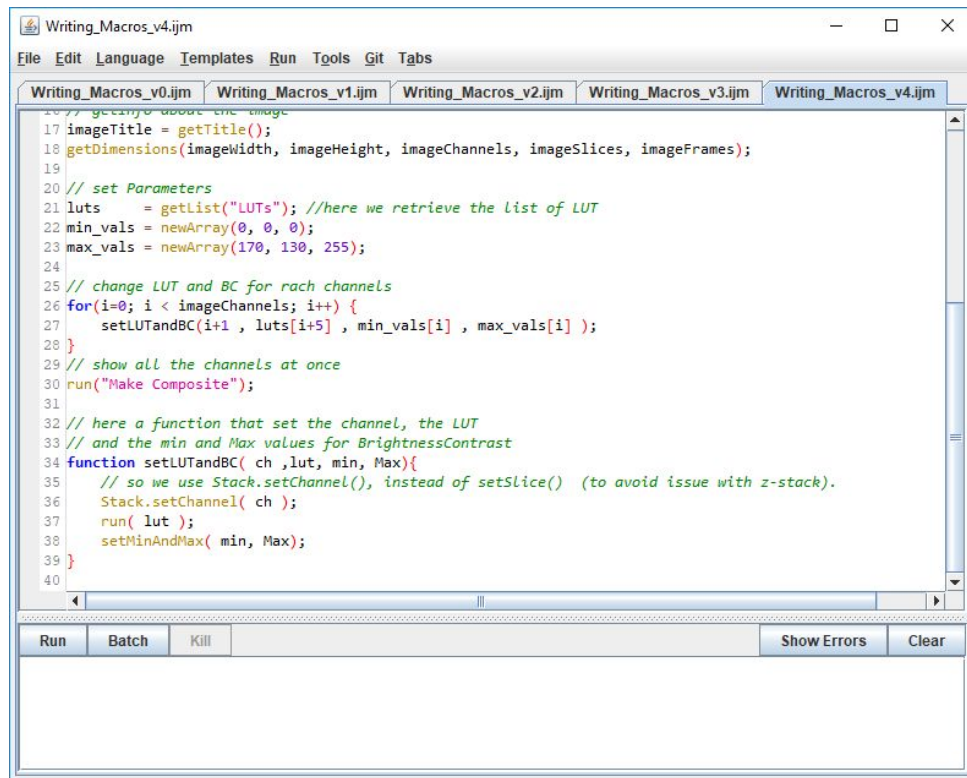
- We get the dimensions of the image to know the number of channels using **getDimensions()**
- We use **Stack.setChannel()** instead of **setSlice()** which will not work in case of hyperstacks.



```
12 * 2018 - Olivier Burri, Romain Guet, EPFL - SV - BIOP
13 * https://biop.epfl.ch
14 */
15
16 // getInfo about the image
17 imageTitle = getTitle();
18 getDimensions(imageWidth, imageHeight, imageChannels, imageSlices, imageFrames);
19
20 // set Parameters
21 luts = newArray("Cyan", "Green", "Magenta");
22 min_vals = newArray(0, 0, 0);
23 max_vals = newArray(170, 130, 255);
24
25 // set the slice, the LUT and the min and Max values for BrightnessContrast
26 // for each channels
27 for(i=0; i<=imageChannels; i++) {
28     // so we use Stack.setChannel(), instead of setSlice() (to avoid issue with z-stack).
29     Stack.setChannel(i+1);
30     run(luts[i]);
31     setMinAndMax(min_vals[i], max_vals[i]);
32 }
33 // show all the channels at once
34 run("Make Composite");
35
36
```

Writing Macros from Scratch

- To encapsulate functionality, we create a function to set the lookup table and display
- It is then used in the loop



```
Writing_Macros_v4.ijm
File Edit Language Templates Run Tools Git Tabs
Writing_Macros_v0.ijm Writing_Macros_v1.ijm Writing_Macros_v2.ijm Writing_Macros_v3.ijm Writing_Macros_v4.ijm
16 // getting info about the image
17 imageTitle = getTitle();
18 getDimensions(imageWidth, imageHeight, imageChannels, imageSlices, imageFrames);
19
20 // set Parameters
21 luts = getList("LUTs"); //here we retrieve the list of LUT
22 min_vals = newArray(0, 0, 0);
23 max_vals = newArray(170, 130, 255);
24
25 // change LUT and BC for each channels
26 for(i=0; i < imageChannels; i++) {
27     setLUTandBC(i+1, luts[i+5], min_vals[i], max_vals[i]);
28 }
29 // show all the channels at once
30 run("Make Composite");
31
32 // here a function that set the channel, the LUT
33 // and the min and Max values for BrightnessContrast
34 function setLUTandBC( ch, lut, min, Max){
35     // so we use Stack.setChannel(), instead of setSlice() (to avoid issue with z-stack).
36     Stack.setChannel( ch );
37     run( lut );
38     setMinAndMax( min, Max);
39 }
40
Run Batch PSIII Show Errors Clear
```